To Sketch Curves by hand we use the following Checklist to get information.

Note, not every step will be used.

A. Domain:

Start by finding the domain of the function so you know what values of x are relevant.

B. Intercepts:

- Find y-intercept by evaluating $f(0)$
- Find x-intercepts by solving $f(x)=0$
  - Can be skipped if calculations are difficult.

C. Symmetry

- Even: If $f(-x)=f(x)$, then f is even and symmetric over x-axis
- Odd: If $f(-x)=-f(x)$, then f is odd and has $180°$ symmetry
- Periodic: If $f(x+P)=f(x)$ for some P, then the function is periodic.

D. Asymptotes:

- **Horizontal:** Where $\lim_{x \to \infty} f(x) = L$ or $\lim_{x \to -\infty} f(x) = L$

- **Vertical:** Values $x = a$ where one of

$$\lim_{x \to a^+} f(x) = \infty, \quad \lim_{x \to a^+} f(x) = -\infty$$
$$\lim_{x \to a^-} f(x) = \infty, \quad \lim_{x \to a^-} f(x) = -\infty$$

E. Intervals of Increasing/decreasing

Compute $f'(x)$ and find where $f'(x) > 0$, and $f'(x) < 0$.

↕ Often done together

F. Local Max and Min

- Find values c where $f'(c) = 0$ or DNE
- Use Part E to decide if c relates to a max or min.

G. Concavity and Points of Inflection

first find $f''(x) = 0$ or DNE

- Find $f''(x)$ and find where $f''(x) > 0$ and $f''(x) < 0$

Concave up — Concave down

- The points concavity changes are points of inflection

H. Sketch the curve

Ex: Sketch $f(x) = \frac{2x^2}{x^2-1}$

A. Domain is $(-\infty,-1)\cup(-1,1)\cup(1,\infty)$

B. y-intercept is $\overset{f(0)}{0}$, x intercept is $\overset{f(x)=0}{0}$

C. $f(-x) = \frac{(-x)^2}{(-x)^2-1} = \frac{x^2}{x^2-1} = f(x)$, So the function is <u>even</u>

D. • $\lim_{x\to\infty} \frac{2x^2}{x^2-1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} = \lim_{x\to\infty} \frac{2}{1-\frac{1}{x^2}} = 2$

• $\lim_{x\to-\infty} \frac{2x^2}{x^2-1} = 2$ as well

• Since denominator is 0 when $x=\pm1$, we check

• $\lim_{x\to1^-} \frac{2x^2}{x^2-1} = -\infty$ • $\lim_{x\to1^+} \frac{2x^2}{x^2-1} = \infty$

if $x\to1^-$, $x<1$
so $x^2-1<0$

• $\lim_{x\to-1^-} \frac{2x^2}{x^2-1} = \infty$ • $\lim_{x\to-1^+} \frac{2x^2}{x^2-1} = -\infty$

Horizontal Asymptote: $y=2$

Vertical Asymptote: $x=-1$, $x=+1$

E. Inc/Dec

$$f'(x) = \frac{(x^2-1)(4x) - (2x^2)(2x)}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$$

$f'(x)=0$ if $x=0$, $f'(x)$ DNE if $x=\pm 1$

	$-4x$	$(x^2-1)^2$	$f'(x)$	Inc/dec
$(-\infty,-1)$	+	+	+	Inc
$(-1,0)$	+	+	+	Inc
$(0,1)$	−	+	−	Dec
$(1,\infty)$	−	+	−	Dec

$$\boxed{\begin{array}{ll} \text{Inc:} & (-\infty,-1)\cup(-1,0) \\ \text{Dec:} & (0,1)\cup(1,\infty) \end{array}}$$

F. from Part E,

$(0,0)$ is local maximum

G. Concavity

$$f''(x) = \frac{(x^2-1)^2(-4) - (-4x)(2(x^2-1))(2x)}{(x^2-1)^4}$$

$$= \frac{12x^2+4}{(x^2-1)^3}$$

$f''(x) = 0$ never $\sim$ $12x^2+4 > 0$

$f''(x)$ DNE if $x = \pm 1$

	$12x^2+4$	$(x^2-1)^3$	$f''(x)$	Concavity
$(-\infty, -1)$	+	+	+	UP
$(-1, 1)$	+	−	−	Down
$(1, \infty)$	+	+	+	UP

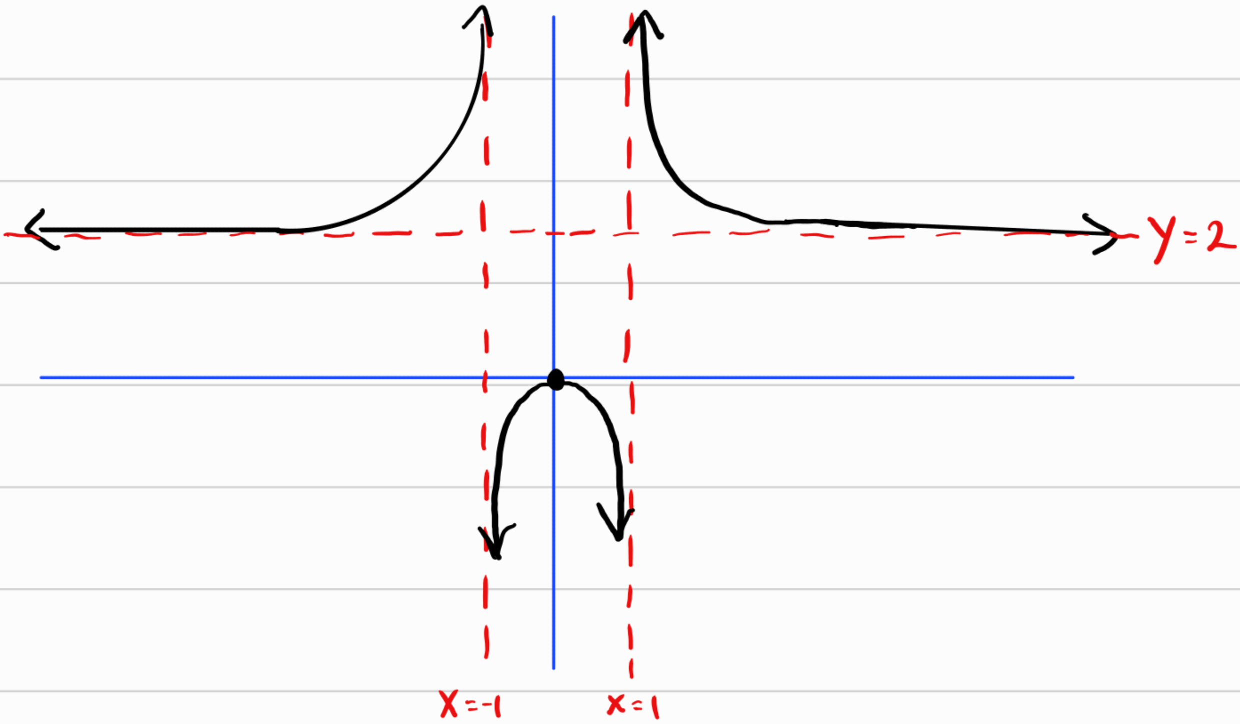
CU: $(-\infty,-1) \cup (1,\infty)$

CD: $(-1,1)$

Points of Inflection

Don't exist as ± 1 aren't in domain of f.

H. Sketch



Ex: Sketch $f(x) = \frac{x^2}{\sqrt{x+1}}$

A. Domain is $(-1,\infty)$

B. x-intercept is 0, y-intercept is 0

C. $f(-x) = \frac{(-x)^2}{\sqrt{-x+1}} = \frac{x^2}{\sqrt{-x+1}} \neq f(x)$ nor $-f(x)$

No Symmetry

D. Asymptotes

$$\lim_{x \to \infty} \frac{x^2}{\sqrt{x+1}} = \infty$$

$$\lim_{x \to -1^+} \frac{x^2}{\sqrt{x+1}} = \infty$$

Horizontal Asymptote: None

Vertical Asymptote: $x=-1$

E. Inc/Dec

$$f'(x) = \frac{(x+1)^{1/2}(2x) - x^2\frac{1}{2}(x+1)^{-1/2}}{(x+1)} = \frac{3x^2+4x}{2(x+1)^{3/2}} = \frac{x(3x+4)}{2(x+1)^{3/2}}$$

$f'(x) = 0$ if $x = 0$ or $x = \frac{-4}{3}$

DNE if $x = -1$

$-\frac{4}{3}$ isn't in domain

	x	$3x+4$	$2(x+1)^{3/2}$	$f'(x)$	
$(-1,0)$	$-$	$+$	$+$	$-$	Dec
$(0,\infty)$	$+$	$+$	$+$	$+$	Inc

Decreasing $(-1,0)$
Increasing $(0,\infty)$

F. Max/Min

f' Changes from negative to positive at 0

So $(0,0)$ is minimum

G. Concavity

$$f''(x) = \frac{2(x+1)^{3/2}(6x+4) - 3(3x^2+4x)(x+1)^{1/2}}{4(x+1)^3} = \frac{3x^2+8x+8}{4(x+1)^{5/2}}$$

$f''(x)$ DNE if $x=-1$

$=0$ if $3x^2+8x+8=0$

Not real as $8^2-4\cdot3\cdot8 = -32$

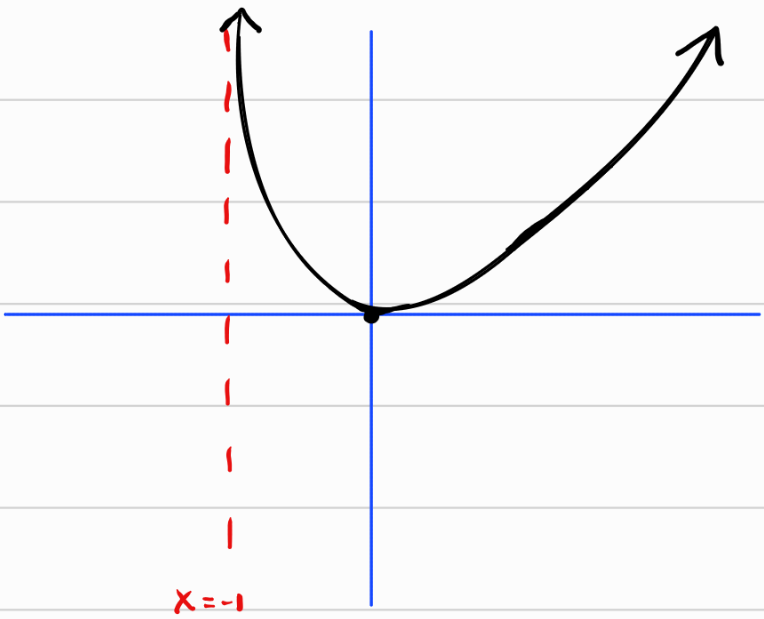
Quadratic formula $x = \dfrac{-8 \pm \sqrt{8^2-4\cdot3\cdot8}}{2\cdot3}$

So $f''(x) \neq 0$

So, there won't be any change in concavity and we check $f''(x)>0$ on $(-1,\infty)$ ~ just check $f''(0)$

So, f is <u>concave up on $(-1,\infty)$</u>

H. Sketch



Ex: Sketch $f(x) = \frac{\cos(x)}{2+\sin(x)}$

A. $2+\sin(x) \neq 0$ So domain is $\mathbb{R}$

B. $f(0) = \frac{1}{2}$, $\cos(x) = 0$ if $x = \frac{\pi}{2} + n\pi$

Y-intercept $\frac{1}{2}$

X-intercept $\frac{\pi}{2} + n\pi$

C. $f(-x) = \frac{\cos(-x)}{2+\sin(-x)} = \frac{\cos(x)}{2-\sin(x)} \neq f(x)$ nor $f(-x)$

However, $f(x+2\pi) = \frac{\cos(x+2\pi)}{2+\sin(x+2\pi)} = \frac{\cos(x)}{2+\sin(x)} = f(x)$

So f is Periodic with period 2π

D. f is Periodic, so $\lim_{x\to\pm\infty} f(x)$ DNE

The denominator is never 0 $(2+\sin(x) \neq 0)$ so there are no Vertical asymptotes

No Asymptotes

E. Inc/dec

$$f'(x) = \frac{(2+\sin(x))(-\sin(x)) - \cos(x)(\cos(x))}{(2+\sin(x))^2} = -\frac{2\sin(x)+1}{(2+\sin(x))^2}$$

$f'(x) = 0$ if $\sin(x) = \frac{-1}{2} \rightarrow x = \frac{7\pi}{6} + 2n\pi, \frac{11\pi}{6} + 2n\pi$

$f'(x)$ always exists.

look at one Period

	-1	$2\sin(x)+1$	$(2+\sin(x))^2$	$f'(x)$
$(0, \frac{7\pi}{6})$	$-$	$+$	$+$	$-$
$(\frac{7\pi}{6}, \frac{11\pi}{6})$	$-$	$-$	$+$	$+$
$(\frac{11\pi}{6}, 2\pi)$	$-$	$+$	$+$	$-$

Inc: $(\frac{7\pi}{6}, \frac{11\pi}{6})$

Dec: $(0, \frac{7\pi}{6}) \cup (\frac{11\pi}{6}, 2\pi)$

F.

Periodic

Max $\left(\frac{11\pi}{6} + 2n\pi, \frac{\sqrt{3}}{3}\right)$

Min $\left(\frac{7\pi}{6} + 2n\pi, \frac{-\sqrt{3}}{3}\right)$

G. Concavity

$$f''(x) = -\frac{2\cos(x)(2+\sin(x))^2 - 2(2\sin(x)+1)(2+\sin(x))(\cos(x))}{(2+\sin(x))^4}$$

$$= -\frac{2\cos(x)(1-\sin(x))}{(2+\sin(x))^3}$$

$f''(x) = 0$ if $\cos(x) = 0$ or $\sin(x) = 1$

$\frac{\pi}{2}, \frac{3\pi}{2}$ $\qquad$ $\frac{\pi}{2}$ — in one Period

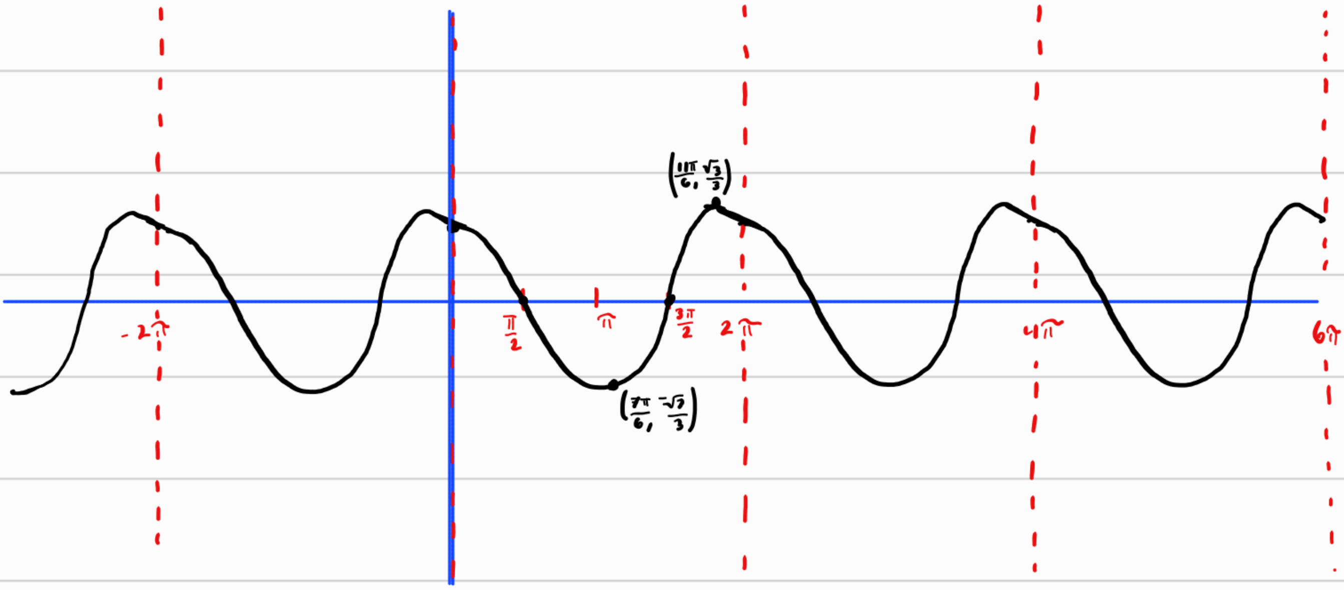
	-1	$2\cos(x)$	$1-\sin(x)$	$(2+\sin(x))^3$	$f''(x)$
$(0, \frac{\pi}{2})$	$-$	$+$	$+$	$+$	$-$
$(\frac{\pi}{2}, \frac{3\pi}{2})$	$-$	$-$	$+$	$+$	$+$
$(\frac{3\pi}{2}, 2\pi)$	$-$	$+$	$+$	$+$	$-$

Concave Up : $(\frac{\pi}{2}, \frac{3\pi}{2})$

Concave Down : $(0, \frac{\pi}{2}) \cup (\frac{3\pi}{2}, 2\pi)$

Point of Inflection: $(\frac{\pi}{2} + 2n\pi, 0), (\frac{3\pi}{2} + 2n\pi, 0)$

H. Sketch



Periodic so image
repeats every 2π